# American Computer Science League

2024-2025 ● Contest 1: Short Problems Solutions ● Classroom Division

**1. Computer Number Systems**

$2024_{10} = 1536 + 448 + 80 + 0$
$= 3 * 8^3 + 7 * 8^2 + 5 * 8^1 + 0 * 8^0$
$= 3750_8$
Also can be done by dividing by 8 and writing the remainders backwards.

C. 3750

**2. Computer Number Systems**

$$\begin{array}{r} 111 \\ 9876_{16} \\ +\ ABC_{16} \\ \hline A332_{16} \end{array}$$

D. A332

**3. Recursive Functions**

$f(1) = 2$
$f(2) = f(2-1) + 2 = f(1) + 2 = 2 + 2 = 4$
$f(3) = f(3-1) + 3 = f(2) + 3 = 4 + 3 = 7$
$f(4) = f(4-1) + 4 = f(3) + 4 = 7 + 4 = 11$
$f(5) = f(5-1) + 5 = f(4) + 5 = 11 + 5 = 16$
The sequence is: 2, 4, 7, 11, 16, 22, 29, 37, 46,...
The difference between terms increases by 1.

C. 16

4. **Recursive Function**

$f(16) = f(16-4) + 1 = f(12) + 1 = 4 + 1 = 5$
$f(12) = f(12-4) + 1 = f(8) + 1 = 3 + 1 = 4$
$f(8) = f(8+1) - 5 = f(9) - 5 = 8 - 5 = 3$
$f(9) = f(9+1) - 5 = f(10) - 5 = 13 - 5 = 8$
$f(10) = f(10-4) + 1 = f(6) + 1 = 12 + 1 = 13$
$f(6) = 2 * 6 = 12$
So $f(f(f(16)) = f(f(5)) = f(10) = 13$ =f $f(19) = 15$

E. 13

**5. What Does This Program Do? (Branching)**

In tracing the program, the last values of x, y, and z are 7, 10, 2.

```
For the program:
   x = 8
   y = 9
   z = -2
   x = y + z
   z = y - x
   y = y + 1
   output x, y, z
```

| x | y | z |
|---|---|---|
| 8 | 9 | -2 |
| 7 | | |
| | | 2 |
| | 10 | |

E. 7, 10, 2

**6. What Does This Program Do? (Branching)**

```
input num
if num % 7 == 0 then
    output num
else
    if num % 5 == 0 then
        output num
    else
        if num >= 60 && num < 80 then
            output num
        end if
    end if
end if
```

Of the numbers 42, 35, 67, 81, 75, 24, 60, 70, 20, 16, the numbers 42, 35, 70 are divisible by 7, the numbers 75, 60, 20 are divisible by 5, and 67 is between 60 and 80. That's 7 of the numbers. Each is output only once.

B. 7

**7. Computer Number Systems**

Convert each hex digit to binary.
$2024_{16}$ = 0010 0000 0010 $0100_2$
Now drop the leading zeros. 10000000100100

B. 10000000100100

| | |
|---|---|
| **8. Computer Number Systems**<br><br>$\begin{array}{r} 111 \\ 4265_8 \\ +\ 6524_8 \\ \hline 13011_8 \end{array}$ | D. 13011 |
| **9. Recursive Functions**<br><br>$f(1) = 256$<br>$f(2) = f(2-1)/2 = f(1)/2 = 256/2 = 128$<br>$f(3) = f(3-1)/2 = f(2)/2 = 128/2 = 64$<br>$f(4) = f(4-1)/2 = f(3)/2 = 64/2 = 32$<br>$f(5) = f(5-1)/2 = f(4)/2 = 32/2 = 16$ | E. 16 |
| **10. Recursive Functions**<br><br>$f(12,10) = f(12-3,10+1) - 2 = f(9,11) - 2 = 8 - 2 = 6$<br>$f(9,11) = f(9+1,11-4) + 1 = f(10,7) + 1 = 7 + 1 = 8$<br>$f(10,7) = f(10-3,7+1) - 2 = f(7,8) - 2 = 9 - 2 = 7$<br>$f(7,8) = f(7+1,8-4+1 = f(8,4) + 1 = 8 + 1 = 9$<br>$f(8,4) = f(8-3,4+1) - 2 = f(5,5) - 2 = 10 - 2 = 8$<br>$f(5,5) = 2 * 5 = 10$ Now substitute backwards. | A. 6 |